

Changsheng Wang

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Research Focus

Trustworthy ML: Machine unlearning, adversarial attack and defense

Generative AI: Diffusion models, large language models (LLMs), multi-modal models

LLM Fine-tuning: Generalization, optimization foundations, downstream task adaptation, reasoning alignment

Education

Michigan State University (MSU)

Sept 2024 – present

Ph.D. in Computer Science

Advisor: [Prof. Sijia Liu](#)

University of Science and Technology of China (USTC)

Sept 2020 – Apr 2024

B.S. in Data Science and Big Data Technology (Rank 5/35)

Industry Experience

Intel, Full-Time, San Jose, USA

May 2025 – August 2025

Research Scientist Intern supervised by [Xin Chen](#) and [Ke Ding](#)

- **Generalization Improvement for the LLM Code Fine-tuning**
 - Identified the *memorization barrier* phenomenon limiting effective fine-tuning.
 - Proposed an information bottleneck–based method to compress spurious memorized features.
 - Improved generalization on large code benchmarks especially for Verilog code.
- **Research Outcome:** Paper under review at **ACL Rolling Review (target: ACL 2026)**.

Research Highlights

1. Data–Optimization Bi-Design for Robust LLM Machine Unlearning

I worked on robust **machine unlearning for LLMs** from multiple perspectives. From the *data perspective*, I examined forget sets through the coreset view [[COLM'25](#)] and the perturbation view [[AISec'25](#)]. From the *optimization perspective*, I applied Invariant Risk Minimization to enhance model robustness against downstream fine-tuning [[ICML'25](#)]. Lastly, we built a robust benchmark to advance community evaluation and understanding of LLM unlearning [[ArXiv'25b](#)].

2. Safety for Large Reasoning Models

I investigated the **safety of large reasoning models** by addressing the leakage of private or harmful reasoning traces. My work [[EMNLP'25](#)] proposes a new paradigm for reasoning-trace unlearning, which adaptively forgets sensitive reasoning patterns while preserving general reasoning capabilities.

2. Fine-Tuning and Downstream Task Adaptation for Code LLM

I studied the **generalization of LLM code fine-tuning** for downstream tasks, focusing on the brittleness of current adaptation pipelines. My work [[ArXiv'25a](#)] introduces the concept of a *memorization barrier*, showing that pretrained models often overfit fine-tuning data already memorized during pretraining. To overcome this, I propose a novel information bottleneck guided fine-tune pipeline that compresses spurious memorized features, reshapes code representations, and enhances generalization on unseen tasks.

Publications

(* Indicates equal contribution)

[[EMNLP'25](#)] **C. Wang***, C. Fan*, Y. Zhang, J. Jia, D. Wei, P. Ram, N. Baracaldo, S. Liu

“Reasoning Model Unlearning: Forgetting Traces, Not Just Answers, While Preserving Reasoning Skills”.

[[ICML'25](#)] **C. Wang**, Y. Zhang, J. Jia, P. Ram, D. Wei, Y. Yao, S. Pal, N. Baracaldo, S. Liu

“Invariance Makes LLM Unlearning Resilient Even to Unanticipated Downstream Fine-Tuning”.

- [COLM'25] **C. Wang***, S. Pal*, J. Diffenderfer, B. Kailkhura, S. Liu
"LLM Unlearning Reveals a Stronger-Than-Expected Coreset Effect in Current Benchmarks".
- [AISec'25] **C. Wang**, Y. Zhang, D. Wei, J. Jia, P.-Y. Chen, S. Liu
"LLM Unlearning on Noisy Forget Sets: A Study of Incomplete, Rewritten, and Watermarked Data".
- [NeurIPS'25] Y. Zhang, **C. Wang**, Y. Chen, C. Fan, J. Jia, S. Liu
"The Fragile Truth of Saliency: Improving LLM Input Attribution via Attention Bias Optimization".
- [WWW'24] W. Wang*, **C. Wang***, F. Feng, W. Shi, D. Ding, T.-S. Chua
"Uplift Modeling for Target User Attacks on Recommender Systems".
- [RecSys'23] **C. Wang**, J. Ye, W. Wang, C. Gao, F. Feng, X. He
"RecAD: Towards A Unified Library for Recommender Attack and Defense".

Preprint Papers

- [ArXiv'25a] **C. Wang**, X. Chen, S. Liu, K. Ding
"Breaking Memorization Barriers in LLM Code Fine-Tuning via Information Bottleneck for Improved Generalization".
- [ArXiv'25b] C. Fan*, **C. Wang***, Y. Huang, S. Pal, S. Liu
"LLM Unlearning Under the Microscope: A Full-Stack View on Methods and Metrics".
- [ArXiv'25c] Y. Lang, Y. Zhang, C. Fan, **C. Wang**, J. Jia, S. Liu
"Downgrade to Upgrade: Optimizer Simplification Enhances Robustness in LLM Unlearning".
- [ArXiv'24] J. Sun, **C. Wang**, J. Wang, Y. Zhang, C. Xiao
"Safeguarding Vision-Language Models Against Patched Visual Prompt Injectors".

Services

Student Activity Chair: The 3rd New Frontiers in Adversarial Machine Learning ([AdvML Frontiers @ NeurIPS2024](#))

Conference Reviewer: NeurIPS'2024, ICASSP'2025, ICLR'2025, ICASSP'2026, ICLR'2026

Honors and Awards

ICML'2025 Travel Grant	2025
USTC Outstanding Graduation Thesis (Top 4%)	2024
USTC Outstanding Students Award (Top 2%)	2021 – 2023
USTC Outstanding Freshman Scholarship (Top 4%)	2020
USTC Robot Game Second Prize (2/16 teams)	2023

Skills

Languages: Python, C, C++, SQL, Matlab, JavaScript

Frameworks: PyTorch, TensorFlow, Keras, Django, Flask

LLM/VLM: LLaMA, Mixtral, LLaVA, Flamingo, DeepSeek, CodeLlama

Tools: LaTeX, MySQL, Docker, Git